

Ubiquitous Sensing, Computing & Communication Laboratory

Quality Laboratory Manual

Experiment No. 04

**To Understand interfacing of Temperature and Humidity Sensor
Using Raspberry Pi 4 Model B.**

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Title of Experiment: To Understand interfacing of Temperature and Humidity Sensor Using Raspberry Pi 4 Model B.

Aim of Experiment: To interface Temperature sensor DHT11 to raspberry pi and measure the Temperature

System Requirements – Raspberry pi 4 Board Win 10 and above OS, 4GB RAM, 2.33 GHz Processor.

Hardware:

- Raspberry Pi 4 Model B
- MicroSD card (with Raspberry Pi OS installed)
- Temperature Sensor DHT11
- Breadboard
- Jumper wires
- Power supply (5V, 3A USB-C adapter)

Software:

- Raspberry Pi OS
- Python 3
- RPi. GPIO library

Experiment Objectives:

Objectives:

1. **Understand the Working Principle of Temperature and Humidity Sensors** – Learn how sensors like **DHT11/DHT22** measure temperature and humidity using resistive and capacitive sensing techniques.
2. **Interface the Sensor with Raspberry Pi 4 Model B** – Establish a proper hardware connection and configure the GPIO pins for data communication.
3. **Implement Data Acquisition using Python** – Write a Python script to read temperature and humidity values from the sensor and display them.
4. **Analyze and Interpret Sensor Readings** – Validate the collected data, observe environmental variations, and understand real-time monitoring applications.

Experiment Outcomes:

- **Successful Interfacing of Temperature and Humidity Sensor with Raspberry Pi** – Properly connect and configure the sensor to read environmental data.
- **Real-time Temperature and Humidity Monitoring** – Collect and display live sensor readings using Python programming.
- **Understanding of GPIO Communication in Raspberry Pi** – Gain hands-on experience in handling digital sensors through Raspberry Pi's GPIO pins.
- **Application in IoT and Embedded Systems** – Develop a foundation for implementing environmental monitoring in smart homes, weather stations, and IoT-based automation projects.

1.Theory:

Temperature and humidity sensors, such as DHT11 and DHT22, are commonly used for environmental monitoring in embedded systems and IoT applications. These sensors measure temperature using a thermistor and humidity using a capacitive humidity sensor. They communicate with microcontrollers like the Raspberry Pi using a single-wire digital signal.

Working Principle

1. **Temperature Measurement**
 - The sensor contains a thermistor (for DHT11) or a temperature-sensitive resistor (for DHT22) to measure temperature based on changes in resistance.
 - The temperature value is converted into a digital signal and sent to the Raspberry Pi.
2. **Humidity Measurement**
 - The sensor uses a capacitive humidity sensor, where changes in air moisture affect the capacitance, allowing humidity calculation.
 - The sensor processes this change and outputs a digital humidity reading.
3. **Data Communication**
 - The sensor operates using a single-wire protocol, requiring only one GPIO pin for data transmission.
 - It sends data in a structured 40-bit format, including temperature, humidity, and checksum values for error detection.

3. Connections

- The VCC pin connects to the 3.3V or 5V supply from the Raspberry Pi.
- The GND pin connects to the GND of the Raspberry Pi.

- The Data pin is connected to a GPIO pin (e.g., GPIO4) with a 10k Ω pull-up resistor to ensure proper signal integrity.

4. Setting up the circuit

We need to install the DHT sensor python library in the Raspberry Pi. but before that Python must be installed in your Raspberry Pi OS. If not installed, then you need to installed by the following commands in the LX Teminal:

```
sudo apt update
```

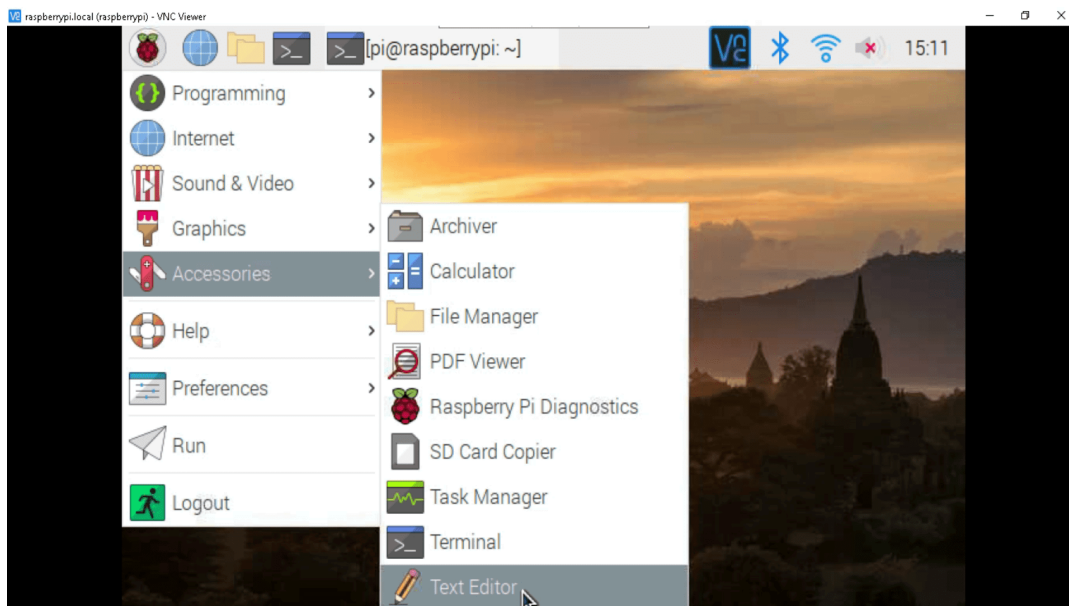
```
sudo apt install python3-pip
```

```
sudo pip3 install --upgrade setuptools
```

Now install the DHT Sensor Pyhton Library:

```
sudo pip3 install Adafruit_DHT
```

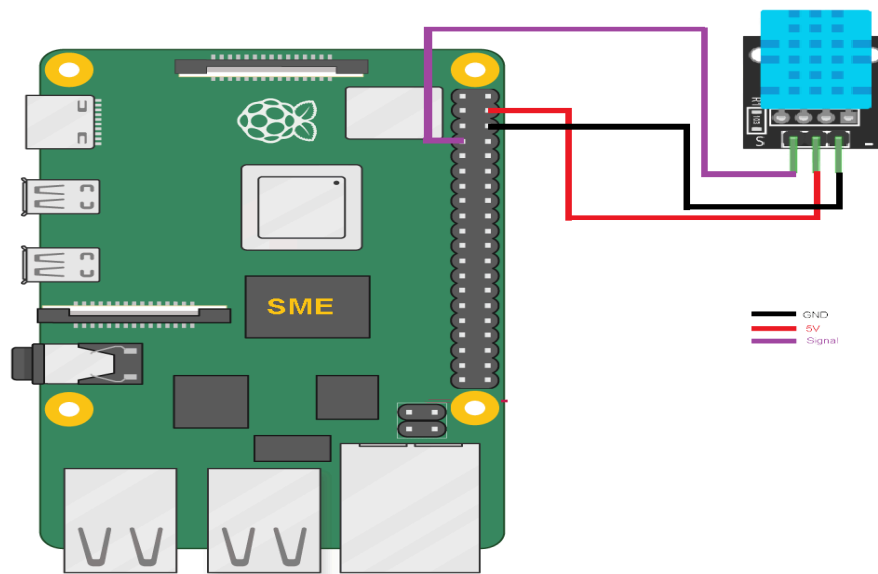
Now, go to the Accessories and open the TextEditor. Copy the below code and paste in to the file.



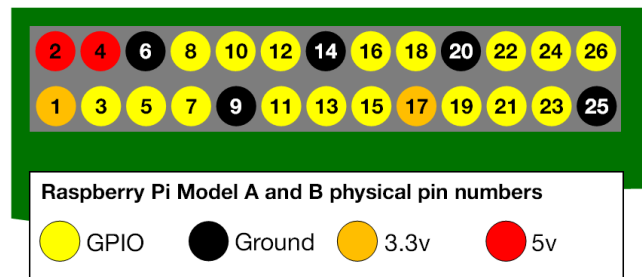
Now save this file in the root folder with name as DHTsensor.py

Now, go to the Accessories and open the TextEditor. Copy the below code and paste in to the file.

Circuit Diagram:-



To find the pin number refer to this diagram showing the physical pin numbers on the Raspberry Pi.



5. Python Code for Ultrasonic Sensor Interfacing with Raspberry Pi

```
import Adafruit_DHT
import time

DHT_SENSOR = Adafruit_DHT.DHT11
DHT_PIN = 4

while True:
    humidity, temperature = Adafruit_DHT.read(DHT_SENSOR, DHT_PIN)
    if humidity is not None and temperature is not None:
        print("Temp={0:0.1f} C Humidity={1:0.1f} %".format(temperature, humidity))
    else:
        print("Sensor failure. Check wiring.");
        time.sleep(2);
```

Now save this file in the root folder with name as DHTsensor.py

Once Saved, open the LX Terminal and Write this command:

Sudo python DHTsensor.py

6.Observations:-

1. **Sensor Response Time Varies** – The **DHT11 sensor** takes about **1 second** to update readings, while **DHT22** provides faster and more accurate data.
2. **Temperature Readings May Slightly Differ** – The sensor may show a slightly higher temperature due to **self-heating** when continuously powered.
3. **Humidity Readings Fluctuate with Environmental Changes** – Rapid changes in humidity (e.g., blowing air or placing a wet cloth nearby) cause noticeable variations in readings.
4. **Stable Readings Require Proper Delay** – Taking readings too frequently may cause incorrect data; adding a **2-second delay** ensures reliable sensor performance.

7.Conclusion:-

The experiment successfully demonstrates the interfacing of a **temperature and humidity sensor (DHT11/DHT22)** with **Raspberry Pi 4 Model B** for real-time environmental monitoring. The sensor provides accurate temperature and humidity readings using a **single-wire digital communication protocol**. Observations indicate that **proper timing and calibration** are essential for stable and accurate data collection. This experiment is useful for **IoT applications, weather monitoring systems, and smart home automation**, making it a valuable addition to embedded systems and sensor-based projects.

Expected Oral Questions:

1. How does the DHT11/DHT22 sensor measure temperature and humidity?
2. Why do we use a pull-up resistor with the data pin of the sensor?
3. What are the key differences between DHT11 and DHT22 sensors?
4. How can the accuracy of temperature and humidity readings be improved in this experiment?

FAQs in Interview:

1. How does the DHT11/DHT22 sensor communicate with the Raspberry Pi, and what type of data format does it use?
2. What are the major differences between DHT11 and DHT22 in terms of accuracy, response time, and operating range?
3. Why is a pull-up resistor necessary for the data pin in the DHT sensor circuit?
4. What are some real-world applications of temperature and humidity sensors in embedded systems and IoT?